CENG 318 / ID 322 - Spring 2026

Bluetooth Earphone Companion App

High-Fidelity & Final Prototype

Joint Assignment II Process Report

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Persona Refinement & Mood Boards

1.1 Introduction to Convergent Stage

Following the divergent thinking approach in the first assignment (where we explored multiple scenarios and basic features), this stage focuses on convergent thinking. We narrow down the interface solutions, clarify interaction needs, and refine our interface design for optimal user experience.

1.2 Refined User Personas

Based on testing and feedback from our low-fidelity designs, we have updated and refined our primary user personas to better reflect their interaction challenges.

1.2.1 Persona 1: Duru Özen (The Creative Professional)

Background: Duru is a 23-year-old photographer living in Istanbul. She leads a dynamic lifestyle where aesthetic expression and creativity are central to both her work and her daily life. She frequently listens to music while editing photos in cafes or studios, requiring fine-tuned audio control.

Core Needs & Pain Points:

- Requires a visually appealing interface that matches her high aesthetic standards as a creative professional.
- Needs precise control over her audio profile (Equalizer) to hear subtle details in the music while working.
- Wants a clear, organized way to map her earbud touch gestures without getting lost in complex menus.

Mapping to High-Fidelity UI: To address Duru's needs, we designed the smartphone app with a premium, dark aesthetic and a clean visual hierarchy. The Custom Equalizer screen provides a 9-band visual curve with intuitive sliders, empowering her to fine-tune her audio. Additionally, the Settings section uses progressive disclosure to keep gesture re-mapping simple and visually clean, preventing cognitive overload.

1.2.2 Persona 2: Duru Özen (The Active Mover)

Background: In addition to her creative work, dancing and running are essential parts of Duru's daily life. Whether she is expressing herself through dance or running along the Bosphorus, she relies heavily on her music but cannot constantly interact with her smartphone.

Core Needs & Pain Points:

- Needs hands-free or highly glanceable controls to adjust volume or skip tracks while dancing or running.
- Finds it disruptive to pull out her smartphone to switch Noise Cancellation modes when moving between quiet streets and loud traffic.
- Requires large, forgiving touch targets on her smartwatch due to the constant movement of her hands and body.

Mapping to High-Fidelity UI: The smartwatch interface was heavily tailored for Duru's active context. The home screen features a massive, glanceable device card that allows her to immediately switch her ANC mode in a single tap. Furthermore, the volume control widget is dominated by a massive 72-point numeric display and large cyclic step buttons (+/-), ensuring she can quickly and accurately adjust her volume while in motion without breaking her rhythm.

1.3 Mood Boards

To strengthen our persona and ensure the digital interface reflects Duru's dynamic lifestyle, we prepared a mood board that captures the intersection of "Creative" and "Sporty" themes. It features vibrant lighting, dynamic movement, and a premium aesthetic that guided our UI color schemes and typography.

Figure 1.1: Mood Board highlighting the "Creative" and "Sporty" aesthetics (dynamic lighting, motion, and modern visuals).

Chapter 2

Experience Mapping & Interaction Flows

2.1 User Experience Maps

This section documents the end-to-end journey of users interacting with the system, highlighting touchpoints, emotional status, and areas of friction.

Table 2.1: Key Experience Map Touchpoints

Stage	User Action	Emotional State	UI Opportunity
Onboarding	Open app, search	Anxious but hopeful	Simple, clear pairing guide
Active Use	Adjusting ANC/EQ	Engaged / Focused	One-tap widgets, immediate feedback
Configuration	Customizing gestures	Analytical	Interactive earbud mapping diagrams

2.2 Refined Interaction Map & Application Flows

We have updated our application flows to support the transition from low-fidelity wireframes to the final high-fidelity working prototypes. The flow integrates smartphone and smartwatch behaviors.

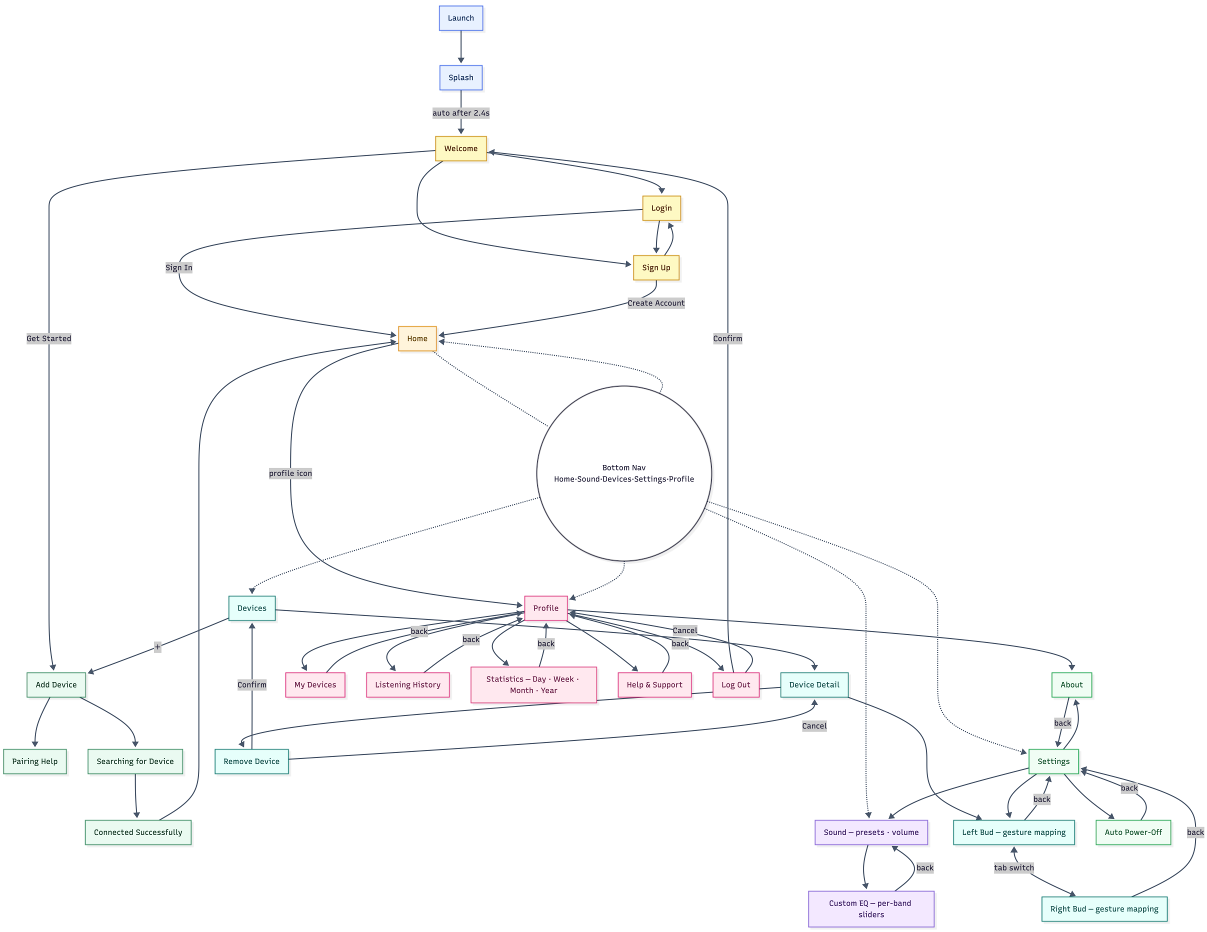

Figure 2.1: Refined Interaction Map and Complete System Navigation Flow.

Chapter 3

Low-Fidelity to High-Fidelity Evolution

3.1 Key Changes & Design Rationale

This section tracks and justifies the changes made between the Version 1/2 wireframes and the final working prototypes.

Table 3.1: Evolution from Low-Fidelity to High-Fidelity

Component / Screen	Low-Fidelity Baseline	High-Fidelity Solution
ANC / Mode Switching	Button matrix on a single, crowded settings page.	Dedicated Home Screen widget with clean visual states (Active/Inactive/Transparency).
Equalizer Controls	Multi-band vertical sliders on the home page.	Simplified 3-band preset controls, expanding to custom sliders with a dedicated "Save Preset" safeguard.
Smartwatch Widget	Non-existent in initial sketches.	Focused circular dial interface showing battery levels, quick ANC toggles, and volume overlay.

3.2 Usability Principles (Norman & Nielsen)

We aligned the high-fidelity UI refinements with standard usability rules to minimize user workload and improve interaction clarity:

- **Visibility of System Status:** Real-time connection feedback is prominently displayed on the Home dashboard with a green "Connected" indicator and individual battery percentage pills (L 100%, R 100%, Case 80%) for clear, immediate status.

- **Match between System and Real World:** The custom Equalizer screen uses a visual multi-band curve with frequency sliders that mimic professional audio mixing hardware, mapping digital sound adjustments to familiar physical expectations.
- **User Control and Freedom:** The interface includes dedicated back navigation buttons ("‹") on all secondary screens, such as the Equalizer and Settings menus, allowing users to safely escape states without getting lost.
- **Consistency and Standards:** Standard bottom tab-bar navigation (Home, Sound, Profile, Settings) is consistently applied across the smartphone app, adhering to standard iOS/Android guidelines for predictable wayfinding.
- **Workload Reduction (Norman's Rules):** The smartwatch application significantly reduces physical workload by introducing a tile-based "Menu" and a quick "Device" shortcut on the home screen, allowing the most frequent action—changing the ANC mode—to be performed in a single glance and tap without requiring the user to open their smartphone.

Chapter 4

Working Interface Proposals

4.1 Working Prototype Access Links

The interactive, high-fidelity prototypes and deployed applications can be accessed and evaluated at:

- **Figma Design Source:** https://www.figma.com/design/pabfCxixRcEJjb6rGrWJxZ/Group16_ID322-CENG318?node-id=2064-2&p=f&t=5MoJ8SxIAHyjkIVK-0
- **Smartphone Prototype (React Deployment):** https://rn-app-eight.vercel.app/
- **Smartwatch Prototype (React Deployment):** https://watch-app-beryl.vercel.app/

4.2 Smartphone Interface Showcase

The smartphone interface is structured across four primary tabs to ensure predictable wayfinding and comfortable one-handed navigation.

4.2.1 Dashboard & Home Screen

The Home Screen acts as the central hub, designed with a dark, premium aesthetic and clear visual hierarchy. At the top, the primary hero card displays the earbud model name, a green "Connected" status indicator, and color-coded battery pills for both earbuds and the charging case, ensuring the most critical information is immediately visible. Below this, prominent toggle buttons are provided for Ambient Sound control (Noise Cancellation, Off, Transparency) alongside a dedicated volume slider. This represents a significant improvement over our low-fidelity wireframes by avoiding a crowded "device control" screen and instead surfacing only the most frequently used settings (status, ANC, and volume) on the top-level view.

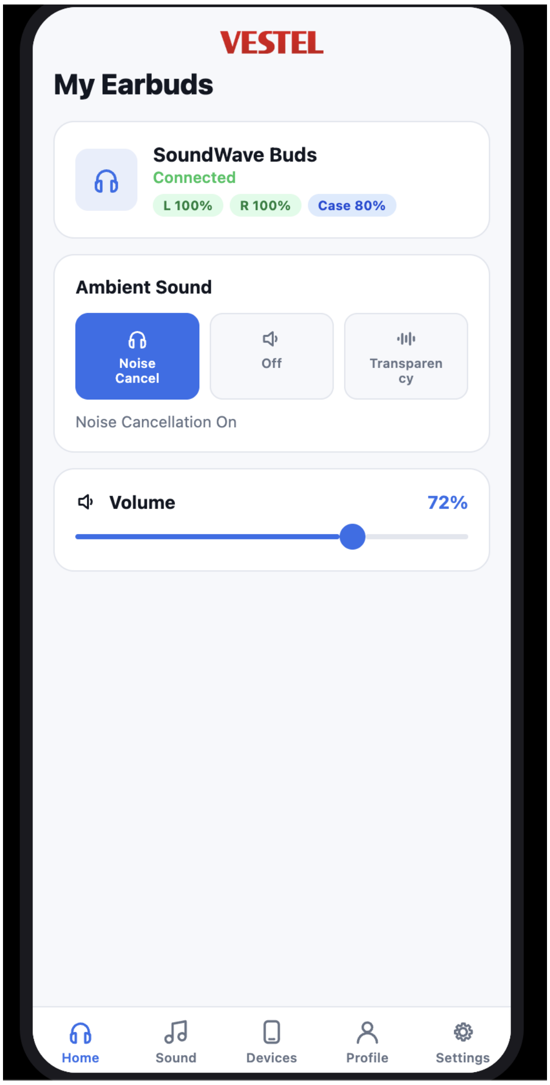

Figure 4.1: High-Fidelity Smartphone Home Dashboard Screen.

4.2.2 Sound Customization & Equalizer

The Custom Equalizer interface enables fine-grained audio tuning. It features an interactive 9-band curve visualization at the top that dynamically reflects adjustments. Below the canvas, simplified sliders are grouped by Bass, Mid, and Treble, rather than overwhelming the user with raw frequency bands. Visual cues are present throughout, with sliders adopting a bright accent color to indicate active manipulation, and clear textual feedback indicating positive or negative decibel increments. A prominent "Save as Preset" button encourages users to safely retain their custom configurations.

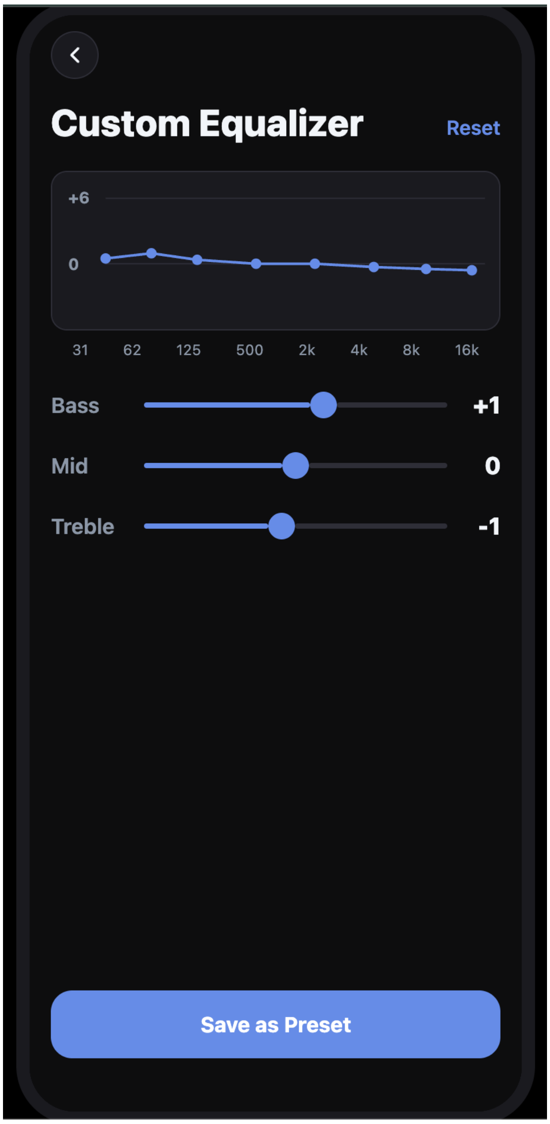

Figure 4.2: High-Fidelity Smartphone Sound & EQ Screen.

4.2.3 Settings & Device Configuration

The Settings and Device Configuration area handles detailed preferences without cluttering the primary tabs. Users can navigate into dedicated screens for Left and Right Earbud gesture mapping (e.g., assigning Next Song or Pause to specific double-taps). The settings are organized hierarchically using progressive disclosure—presenting simple list items initially that only reveal deeper configuration dialogs when tapped. This structure empowers the user to customize their experience safely without feeling overwhelmed by an extensive, continuous list of options.

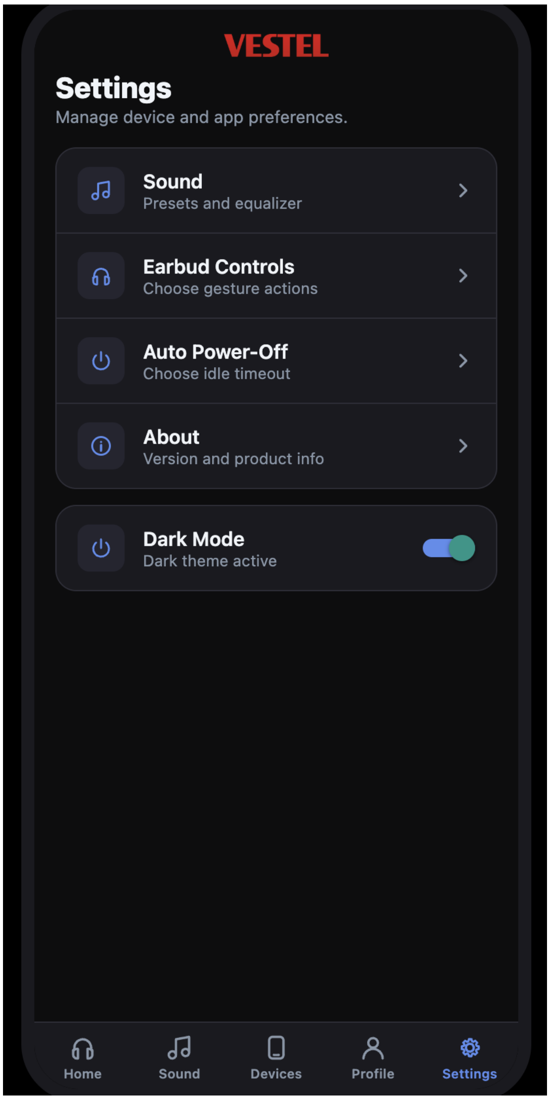

Figure 4.3: High-Fidelity Smartphone Settings Screen.

4.3 Smartwatch Interface Showcase

As required by the assignment guidelines, the smartwatch UI is documented here in its own dedicated section. The design emphasizes immediate glanceability and easy touch targets for small screens.

4.3.1 Smartwatch Home & Quick Controls

The Smartwatch Home interface is tailored for extreme glanceability and constrained, squircle displays. At its core, it features a primary device card that immediately communicates the connected earbud name and current noise control mode. The layout utilizes a prominent grid of large, tap-friendly tiles, specifically separating the "Menu" and "Device" shortcuts. To accommodate the small screen size, text labels are kept minimal and accompanied by distinct, high-contrast iconography, ensuring users can tap accurately even while moving or exercising.

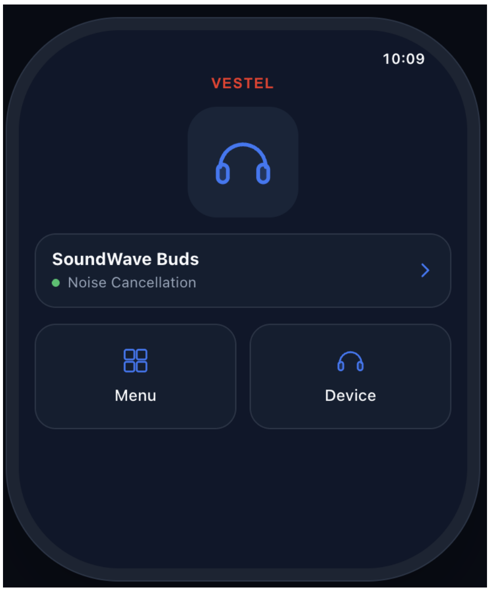

Figure 4.4: High-Fidelity Smartwatch Home Controls Screen.

4.3.2 Smartwatch Volume Control Widget

The Smartwatch Volume interface centers around a massive, 72-point numeric display that dominates the screen, providing instant visual feedback. Users adjust the volume through a dedicated control row positioned at the bottom, which supports cyclic step increments (e.g., adjusting by 1, 5, or 10 steps at a time). This setup allows the user to either quickly tap the plus or minus buttons or hold them for rapid adjustments, ensuring the volume can be confidently managed without taking focus away from an ongoing activity.

Figure 4.5: High-Fidelity Smartwatch Volume Interface.

4.4 Interface Layout, Spacing, and Padding Guidelines

Consistent spacing and padding ensure the layouts are touch-friendly and visually balanced.

- **Touch Targets:** All interactive elements are at least 48 dp $\times$ 48 dp to prevent miss-taps.
- **Screen Padding:** A minimum 16 dp outer safety margin from the smartphone screen borders.
- **Smartwatch Padding:** A minimum 8% inset margin to accommodate curved screen corners on circular and squircle watch face displays.

Chapter 5

Industrial Design & Physical Integration

5.1 Physical Product & Earphone Casing Proposal

This section represents the physical design work developed by the Industrial Design (ID) students, ensuring consistency between the hardware and software experience.

Figure 5.1: Industrial Design: Earphone and Charging Case Rendering.

5.2 Storyboard for Physical Use & Interaction

The storyboard illustrates Duru's interaction with the product during a typical run. It highlights how she seamlessly transitions from using the smartphone app for initial setup to relying entirely on the smartwatch interface and earbud touch gestures to adjust volume and control playback while staying active and focused on her workout.

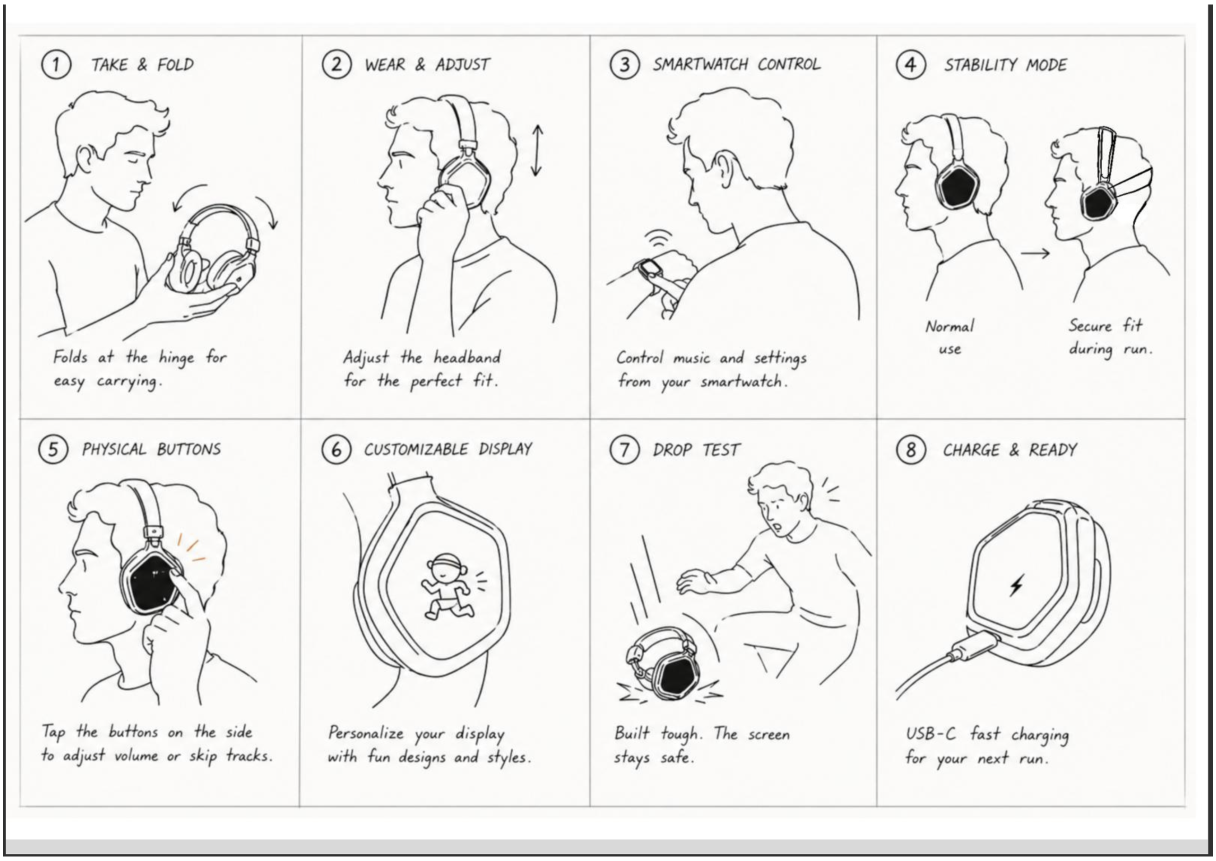

Figure 5.2: Storyboard: Everyday User-Product-App Interaction Journey.

Chapter 6

Conclusion

This process report summarizes the transition from our early exploratory concepts to a coherent, usability-tested high-fidelity interface and physical prototype for both smartphone and smartwatch platforms. By aligning with Nielsen and Norman's usability principles, implementing proper touch padding, and seamlessly integrating the physical earphone design with the digital UI, we have achieved a refined and unified product experience as a collaborative, cross-disciplinary team.